

Localise the red eye

Pain, visual loss and corneal or pupil abnormalities separate urgent disease from uncomplicated surface inflammation.

From symptom to structure

Pattern	Likely site
Discharge with preserved vision	Conjunctiva: viral, bacterial or allergic disease
Deep boring pain; movement worsens pain	Sclera; assess for nonblanching violaceous redness
Photophobia, blur and corneal opacity	Cornea; look for an epithelial defect or infiltrate
Cells, flare or abnormal pupil	Anterior chamber inflammation or acute pressure rise

Examine before reassuring

Record visual acuity, pupils and pressure, then inspect lids, cornea and anterior chamber. Evert the lid when discharge or allergy is suspected. Severe pain, reduced vision, corneal opacity, a shallow chamber or abnormal pupil requires urgent ophthalmic assessment. Ask about contact lenses, infectious contacts and joint or back symptoms: these clues help explain the anatomical findings.

Recognise a corneal ulcer

A painful corneal opacity with an overlying epithelial defect is treated as a corneal ulcer until assessed.



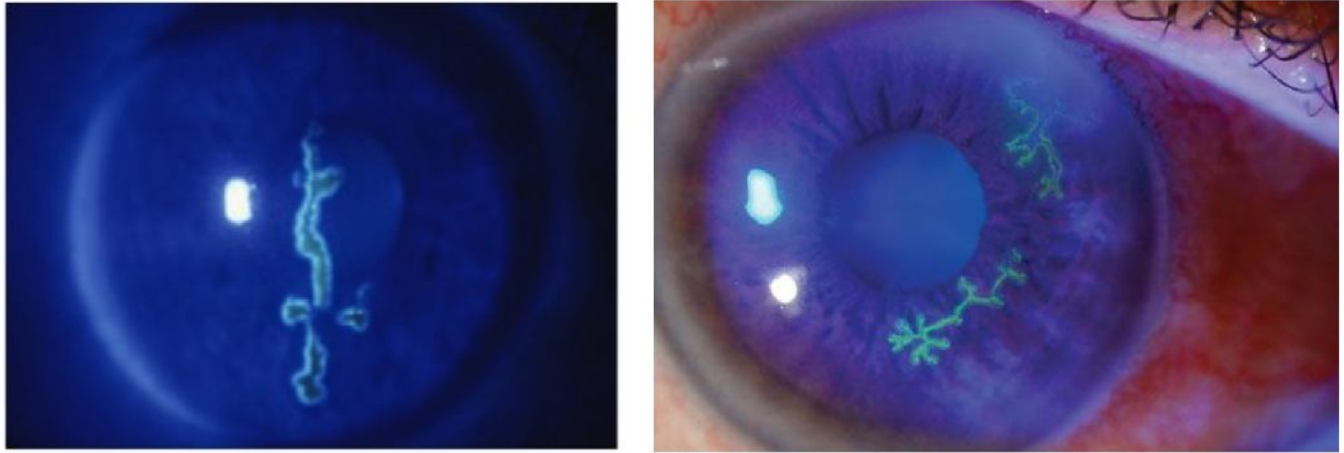
A dense central corneal infiltrate with surrounding haze and a red eye. The photograph demonstrates substantial corneal disease, rather than the clear cornea expected in uncomplicated conjunctivitis; it does not identify the pathogen by appearance alone.

Sample, cover and reassess

Contact lens wear particularly raises concern for *Pseudomonas*. Ophthalmology obtains corneal scrapings for Gram stain and culture, then starts intensive empirical broad-spectrum topical antibiotics and modifies treatment according to sensitivity. Stop contact lens wear. Severe infection, the only functioning eye or difficulty administering frequent drops favours admission. Hypopyon reflects marked inflammation; thinning, perforation and endophthalmitis are major threats. Review the clinical response closely rather than assuming that any initial drop is sufficient.

Separate viral and fungal patterns

The history and corneal lesion determine whether antibacterial therapy alone is appropriate.



Dendritic ulcers (under fluorescein stain + cobalt blue filter)

Fluorescein-stained branching epithelial ulcers under cobalt blue illumination. Terminal bulbs, recurrence and reduced corneal sensation support epithelial HSV keratitis. The original paired photograph retains its teaching label; do not infer laterality from this image.

Match therapy to the process

Epithelial HSV involves active viral replication: treat with topical antiviral and avoid steroid alone. Stromal HSV has an inflammatory component and may need supervised steroid treatment with antiviral cover. HZO combines a V1 vesicular rash with ocular disease; nasal-tip involvement signals increased ocular risk. Organic-matter trauma, feathery margins and satellite lesions suggest fungal keratitis. Obtain appropriate corneal sampling; filamentous fungi such as *Fusarium* are treated with natamycin. Antibacterial treatment alone does not address these nonbacterial causes.

Anterior inflammation and its consequences

Anterior chamber cells establish inflammation; complications can arise from the cornea through to the macula.

Read the slit-lamp findings

Finding	Meaning
Cells and flare	Leukocytes and protein enter aqueous after barrier breakdown
Keratic precipitates	Inflammatory deposits on corneal endothelium
Posterior synechiae	Iris adheres to lens; pupil block may follow
Raised IOP or central blur	Check drainage obstruction and macular oedema

Treat and monitor

AAU typically produces pain, photophobia and blur with little discharge. Topical steroids control inflammation; supervised tapering and follow-up reduce recurrence and identify cataract or pressure complications. Cycloplegics relieve ciliary spasm and help prevent synechiae. Ask about inflammatory back pain, arthritis, skin and urinary symptoms, then target systemic investigations. Painless floaters with predominant vitreous inflammation suggest intermediate uveitis. Persistent central distortion warrants assessment for cystoid macular oedema.